

Designed by Remus Teodorescu 2026

LECTURE 10.2

Solid Oxide Cells

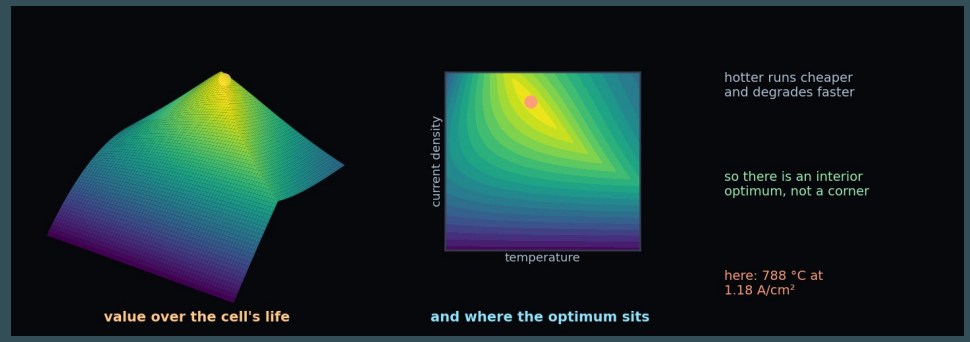
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

STEAM · VOLTAGE · LIFETIME



Schematic: the value a cell produces over its lifetime against temperature and current, with its best point in the interior - hotter is cheaper per hour and ages faster. The lecture sets out the electrolyser and the thermoneutral voltage, solves the fuel channel, and finds the best voltage by a gradient through one PINN trained over the whole range of voltages.

Learning Objectives

SECTIONS, IN ORDER

1. **The cell** — three electrolysers, the losses, the thermoneutral voltage

2. **The channel** — steam running short, a current that follows the voltage

3. **The best voltage** — one PINN for every voltage, the worth and its gradient

Incropera carries the transport, Liu the optimisation through a network. The example is Ex_10.2's electrolysis cell: the channel of L9.2, now with its electrode. This is the last physics lecture, and the place Part 2 has been heading.

READ, AND WHAT EACH ANSWERS

Incropera ch. 1	where do the losses sit?
Incropera ch. 14	how does the steam run short?
Liu ch. 5	what does a gradient buy?

Three Electrolysers

Water electrolyser — a cell that splits water into hydrogen and oxygen with electricity. The three industrial types differ in their electrolyte, and so in the ion that carries the charge and in the temperature at which they run.

ONE ELECTROLYTE EACH

alkaline — a potassium hydroxide solution near 80 degrees; OH^- carries the charge and a diaphragm keeps the gases apart

PEM — a humid polymer membrane near 70 degrees; H^+ crosses to the hydrogen side and liquid water is split at the oxygen side

solid oxide — a ceramic near 800 degrees; O^{2-} crosses to the air side, steam is split and heat supplies part of the energy

COST AND CONTROLLABILITY

	alkaline	PEM	solid oxide
temperature [°C]	70 – 90	50 – 80	700 – 850
energy [kWh/kg H_2]	50 – 78	50 – 83	45 – 55
cost [USD/kW]	500 – 1000	700 – 1400	stack >2000
minimum load [%]	10 – 30	10 – 40	about 50
ramp [% per s]	0.5 – 5	3 – 10	about 0.2
start from cold	minutes	minutes	hours: kept hot

Typical values, rounded. The costs are IRENA's 2020 system prices; installed plants cost more: 2000 – 2600 USD/kW outside China in 2024, 600 – 1200 USD/kW in China (IEA, 2025).

IRENA (2020); Dorn et al. (2026); IEA (2025)

Electrolysis at Industrial Scale

Power-to-X — the conversion of renewable electricity into hydrogen by electrolysis, and of the hydrogen into a fuel or chemical, such as methanol, that can be stored and shipped.

KASSØ, DENMARK, SINCE 2025

electrolysers	52 MW of PEM, from Siemens Energy
power	a 304 MW solar park beside the plant
product	up to 42 000 t of e-methanol a year

INSTALLED WORLDWIDE, AND 2030

installed	0.6 GW in 2021, 2 GW in 2024, 4.2 GW in 2025
by technology	in 2022: alkaline 60 %, PEM 30 %, SOEC under 1 %
by 2030	26 GW committed; 520 GW announced, much now out of reach
China	65 % of the capacity installed or committed



Laura Mærsk, e-methanol fuel
Jens Dohrn, CC BY-SA 4.0



2 MW PEM module, ITM Power
Bubble60, CC BY-SA 4.0

WHY ALKALINE STILL LEADS

mature	a century in industry
cheap	no noble metals, large stacks
China's makers	alkaline at 600 – 1200 USD/kW

Elsewhere 2000 – 2600 USD/kW (IEA). More: Kassø, ITM Power.

The Solid Oxide Electrolyser

Solid oxide electrolysis cell — a ceramic electrochemical cell that splits steam into hydrogen and oxygen near 800 degrees. The name comes from its electrolyte, a solid oxide (yttria-stabilised zirconia) that conducts only O^{2-} ions. In reverse it is a solid oxide fuel cell; the electrodes are therefore the fuel and air electrode, as anode and cathode swap between modes.

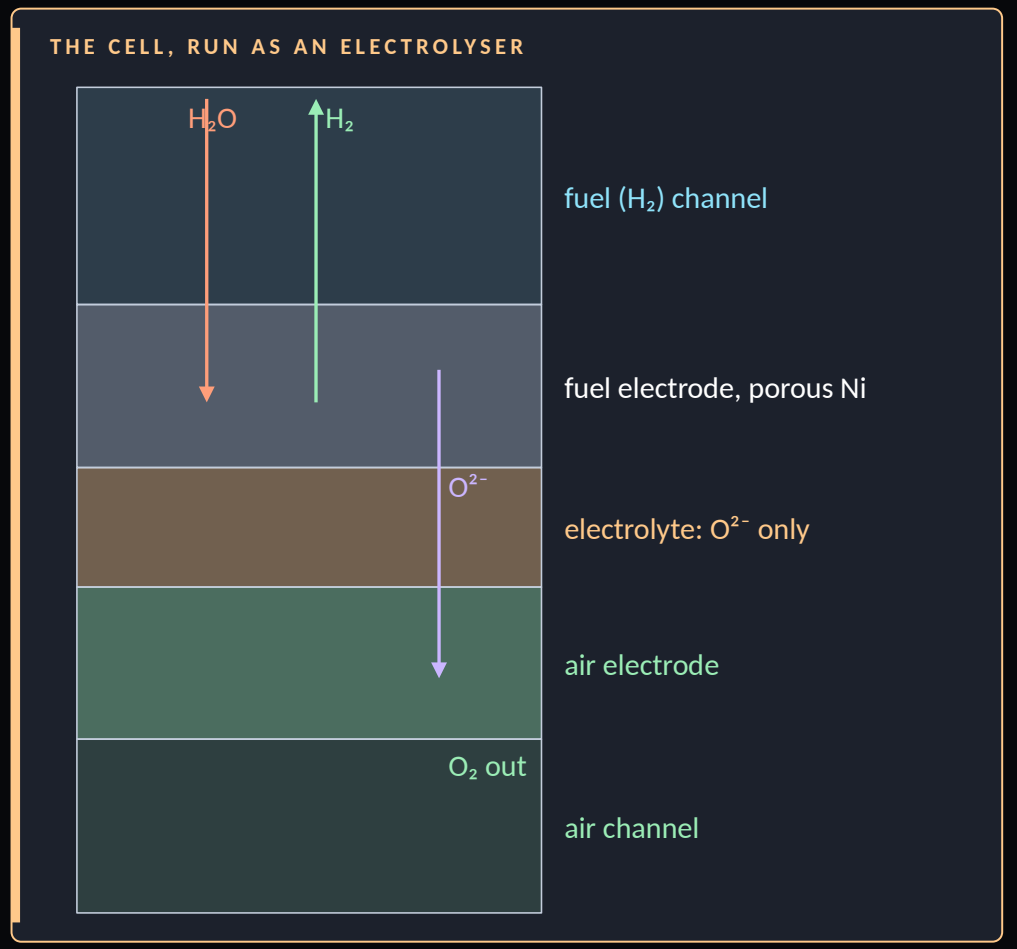
WHY THE HIGH TEMPERATURE PAYS

$$\Delta H = \Delta G + T\Delta S, \qquad \Delta G \downarrow \text{ as } T \uparrow$$

ΔH the energy to split water, ΔG the part that must be electricity, $T\Delta S$ the part heat can supply

THE THREE LAYERS

fuel electrode	steam takes two electrons, hydrogen leaves
electrolyte	carries O^{2-}, keeps the gases apart
air electrode	O^{2-} gives up its electrons as oxygen



The Polarisation Curve

Nernst potential — the cell voltage at which no current flows, the reversible voltage of the reaction at the local gas composition; it rises as the steam is used up. The polarisation curve is this potential plus the losses, which grow with the current density.

THE NERNST POTENTIAL, AND THE LOSSES ON TOP

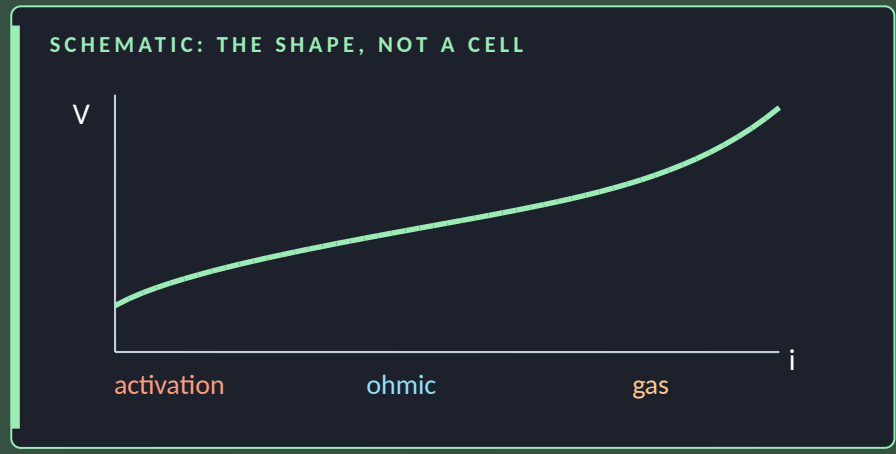
$$E = E^0(T) + \frac{RT}{2F}\ln\left(\frac{(1-y)\,p_{O_2}^{1/2}}{y}\right)$$

$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

E the Nernst potential, y the steam fraction, p_{O_2} the air side's oxygen, V the cell voltage η_{act}, η_{ohm}, η_{conc} the activation, ohmic and gas-transport losses; in electrolysis they add

The exercise gathers all three losses into one area-specific resistance, 0.25 Ω cm², an estimate that stands for the curve at its operating point.

THE SHAPE OF THE CURVE



electrolysis the losses add to it

in an SOC the ohmic loss leads

Button-cell curves are the validation data.

The Thermoneutral Voltage

Thermoneutral voltage — the operating voltage at which the heat the losses release equals the heat the reaction absorbs, so the net heat is zero and the temperature stays even. For steam near 800 degrees it is 1.29 volts, and stacks are designed at or just above it.

THE BALANCE, AND THE HEAT IN ONE TERM

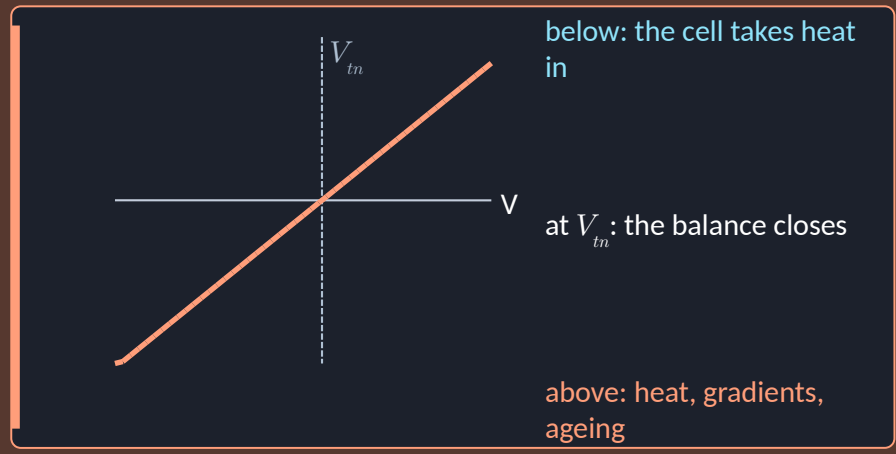
$$V_{tn} = \frac{\Delta H}{2F} = 1.29\ \text{V} \quad (\text{steam},\ 800\ ^\circ\text{C})$$

$$q = i(V - V_{tn})$$

V_{tn} the thermoneutral voltage, ΔH the enthalpy of splitting, q the heat per electrode area, i the current density

TALK · 2 MIN How would you explain, in your own words, where the heat of an electrolyser comes from and where it goes? Then, in the drawing on the right, what does running just above the dashed line cost, and what does it buy?

THE HEAT AGAINST THE VOLTAGE



one clean term all the losses gather in q

the catch ASR falls as T rises

A hotter spot draws more current, and heats further.

The Fuel Channel

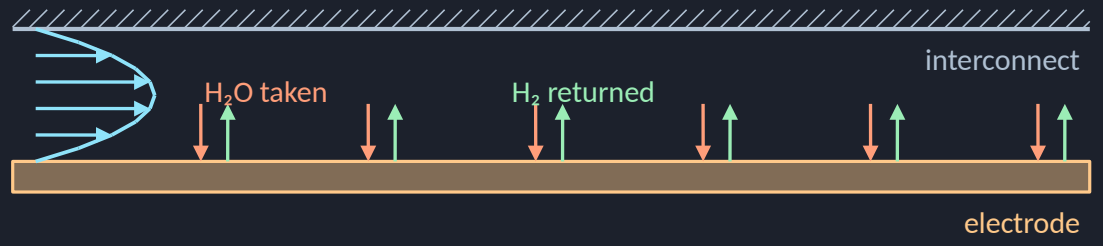
Fuel channel — the hydrogen channel of L9.2, run as an electrolyser: 90 % steam enters at 1 m/s, the electrode takes steam and returns hydrogen, and the steam fraction falls along the channel. The flow is the same known Poiseuille; what changes is the source, which is no longer a given flux.

THE SCALED EQUATION OF EX_10.2

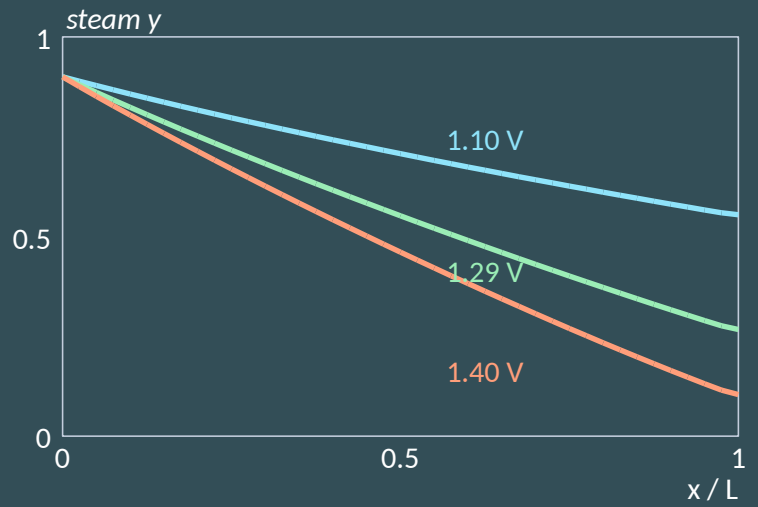
$$y_x = \frac{y_{xx}}{Pe} - K(V - E(y)), \qquad Pe = 125, \quad K = 1.83\ \mathrm{V}^{-1}$$

y the steam fraction, x along the channel over L, Pe the Péclet number, K gathers L, u, h, c, F, ASR

STEAM TAKEN, HYDROGEN RETURNED



STEAM ALONG THE CHANNEL



The exercise's reference at three cell voltages. At 1.29 V the steam fraction at the outlet is 0.27, so 69 % of the steam is used; at higher voltage the outlet starves.

The Local Current

Current from the voltage — the line that changes everything from the last lecture: the wall flux is no longer given. The cell sits at one voltage, the local gas sets the local Nernst potential, and their gap over the resistance is the local current - highest where the steam is fresh.

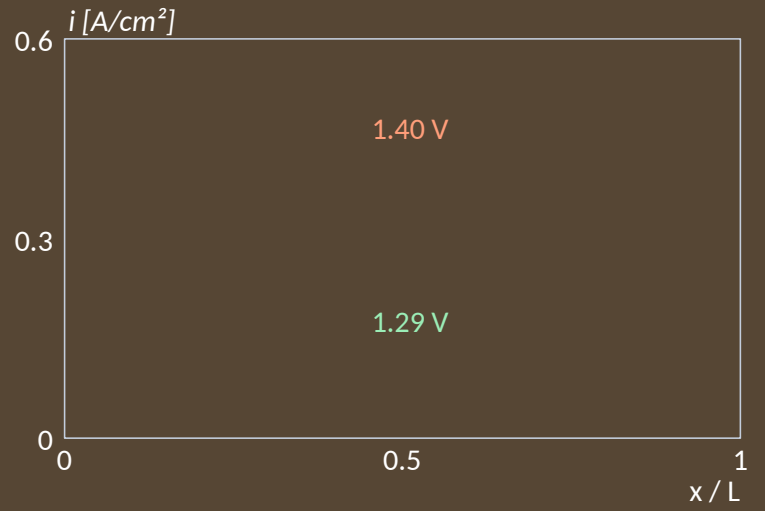
ONE VOLTAGE, A CURRENT THAT VARIES

$$i(y) = \frac{V - E(y)}{ASR}$$

i the local current density, $E(y)$ the Nernst potential of the local gas, ASR all losses as one area resistance (an estimate)

So the equation is nonlinear in y through $E(y)$, and the outlet needs a condition again: the exercise holds the slope at zero there with a loss term, and solves the reference with a boundary-value solver rather than a march.

THE CURRENT ALONG THE CHANNEL



Highest at the inlet, where the steam is fresh. A starved outlet runs a higher local loss and ages faster: the cell fails at one end first.

One PINN for Every Voltage

Parametric PINN — the move of the hydrogen channel again: the cell voltage joins the position as an input, and after one training over 1.10 to 1.40 V any voltage is read off, not solved again. The inlet is built into the trial function; the zero-slope outlet is a loss term.

THE TRIAL FUNCTION, AND ITS RESIDUAL

$$\hat{y}(x, V) = y_{in} + x\,\mathcal{N}(x, V)$$

$y_{in} = 0.90$ the inlet, built in, $\mathcal{N}$ the network: x and the cell voltage V in, trained over 1.10 to 1.40 V

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4)      # x and the cell voltage V in
y = Y_IN + x * net(p)                           # the trial function: the inlet
built in
g = grad(y, p)                                  # first derivatives by autograd
res = g[:,0:1] - d2(y, p, 0) / PE + K * (V - nernst(y))   # the residual
out = grad(y, p)[outlet, 0:1]                   # the outlet slope, held at 0
```

THREE METHODS AT 1.29 V

	FVM, 50 cells	PINN
worst error	0.0042	0.0005
hydrogen	5.20 g/h	5.21 g/h
one solution	2 ms	under 1 ms
a new voltage	solve again	read off
training, offline	none	20 s

The reference is a boundary-value solver at tolerance 10^{-10}. The same network is within 0.0013 of it at 1.10, 1.20, 1.35 and 1.40 V: one training, the whole range.

The Worth of an Hour

Objective — what the operation is for, written as one number: the hydrogen made in an hour, at its price, less the electricity used, at its price. The physics supplies the two rates; the prices are assumptions, not physics, and the result is conditional on them.

THE OBJECTIVE, AND ITS GRADIENT

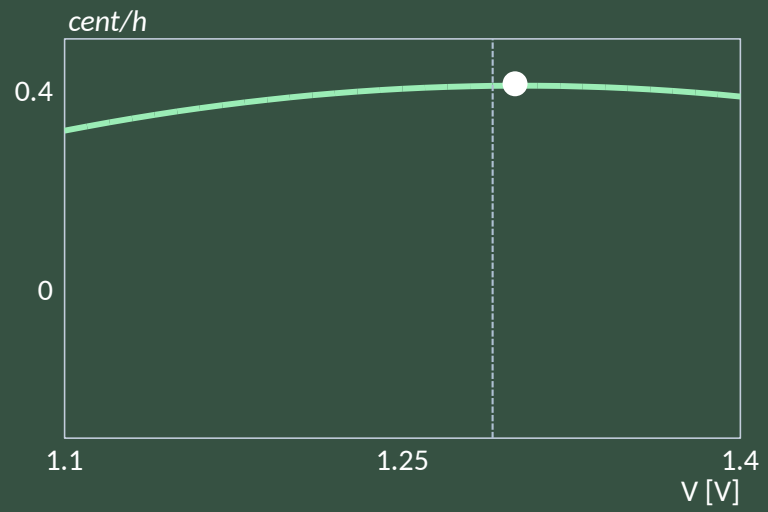
$$J(V) = p_{H_2}\,\dot{m}_{H_2}(V) - p_{el}\,P(V), \qquad \frac{dJ}{dV} \text{ by autograd}$$

J the worth of an hour, p_{H_2}, p_{el} the prices (assumptions, not physics), $\dot{m}_{H_2}$ hydrogen made, P power drawn

THE BEST VOLTAGE IN PYTORCH

```
V = torch.tensor([1.25], requires_grad=True)   # the knob, differentiable
for k in range(200):
    J = value_per_hour(net, V)                 # hydrogen minus electricity
    V = V + lr * grad(J, V)                    # uphill, through the physics
```

CENT PER HOUR AGAINST VOLTAGE



by the gradient	1.30 V, 200 steps, 0.2 s
by 61 solver runs	1.30 V, 4 s

No Best Point Over a Lifetime

Degradation — the slow rise of the cell's resistance: nickel coarsens, interfaces delaminate, chromium poisons the air electrode. The rate is Arrhenius in T and a power of i, so what produces more now ages faster - and published rates span an order of magnitude.

THE DEGRADATION LAW, AND THE LIFETIME OBJECTIVE

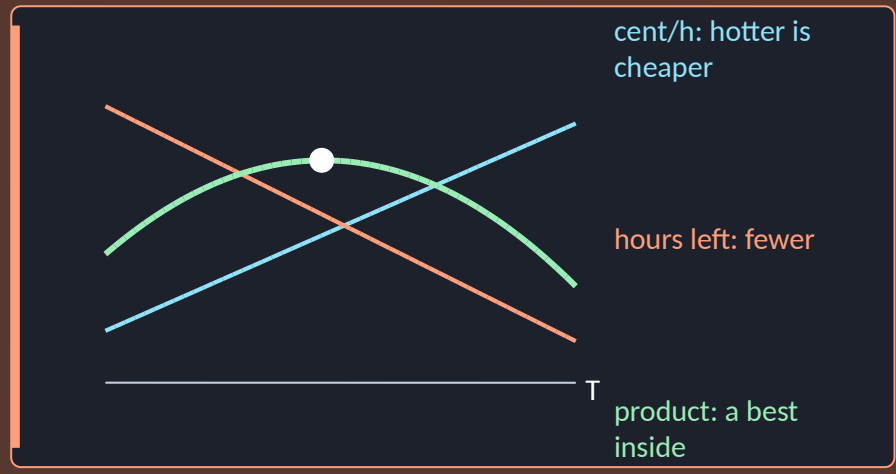
$$\frac{dASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right) |i|^n$$

$$\max_{V(t),\,T(t)} \int_0^{t_{life}} [p_{H_2}\dot{m}_{H_2} - p_{el}P]dt \quad \text{s.t.} \quad ASR(t_{life}) \le ASR_{\max}, \quad |\nabla T| \le |\nabla T|_{\max}$$

k_0, E_d, n the degradation law's constants : Arrhenius in T, a power of i; published rates span an order of magnitude

TALK · 2 MIN How would you explain, in your own words, why a device like this has no best operating point? Then, in the drawing on the right, what moves the white dot when the price of electricity falls?

THE TRADE, DRAWN



so the answer a trajectory $V(t), T(t)$

held back by lifetime, and $|\nabla T|$: ceramic

The Gradient Through the Physics

Differentiable model — a model that gives the fields and their derivatives with respect to anything, by the autograd that trained it. A solver evaluates the physics; optimising through it costs a solve per variable per step, or an adjoint. Through a PINN the gradient is read off.

WHAT EACH PREDICTED FIELD SUPPLIES

the field	from	what it supplies here
temperature	L8	the $\lvert\nabla T\rvert$ limit the ceramic sets
composition	L9	the local Nernst potential
current, potential	L10	the degradation it drives

Part 2's argument in one table: none of those fields was an end in itself. They are inputs to a decision, and constraints are written in terms of them.

SOLVER AGAINST NETWORK, OPTIMISING

solver	one solve per variable per step
or	an adjoint, built by hand
network	the gradient from autograd
in Ex_10.2	0.2 s against 4 s of sweeps
to refuse	a lifetime as a promise

Degradation rates are uncertain by an order of magnitude: a predicted lifetime ranks schedules, nothing more.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
the channel with its temperature	the cell cools below V_{tn}; current follows	Incropera ch. 5	stacks; MP10.2A
the degradation law from records	k_0, E_d, n fitted to long voltage drifts	Ex_10.2 truth	lifetime models; MP10.2B
the button-cell fit	kinetics and resistance against measured curves	Incropera ch. 1	validation before a channel
the daily schedule	the hourly electricity price as the driver	Liu ch. 5	grid-coupled operation; L12
the steam supply and its heat	exchangers that make the gain real	Incropera ch. 11	plant efficiency

WHY THEY ARE LEFT OUT

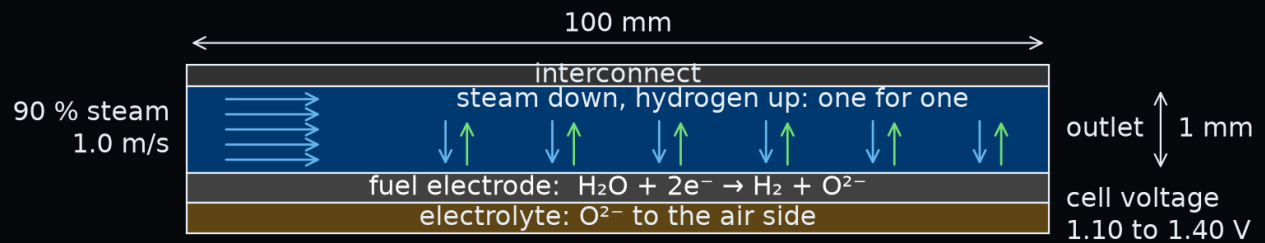
Ex_10.2 solves one isothermal channel at one moment, scored against a boundary-value solver, so none of them is needed to do it. The first two rows are the two mini projects; the last two are where the schedule meets the grid of L12.

EXERCISE

Exercise 10.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the electrolyser, its channel, the Nernst potential and K
- **4** — the boundary-value reference; finite volumes with Newton
- **5** — one PINN for every voltage from 1.10 to 1.40 V
- **6** — the best cell voltage, by a gradient through the network
- **7 – 9** — the report; mini projects MP10.2A and MP10.2B



THE QUESTION EACH PART ANSWERS

1 – 3	what sets the local current?
4	how fine must the grid be?
5	one network, every voltage?
6	which voltage pays best?

REPORT QUESTION, TO CONCLUDE

The ASR, the diffusivity and the prices are the notebook's estimates, and two students with different price ratios should find different best voltages. What in your answer is physics, and what is an assumption?

SUMMARY

Key Takeaways

SOLID OXIDE CELL

what it is	a ceramic cell, two directions
function	steam to hydrogen, heat helping
train, tune	run at or just above V_{tn}
vs low temperature	a quarter less electricity
used for	storing surplus power as fuel

LIFETIME OPTIMISATION

what it is	a PINN with the voltage as input
function	the worth, and its gradient
train, tune	once, over 1.10 to 1.40 V
vs a solver	the gradient comes free
used for	a trajectory, not a setting

Next — L11.1: a network in a real-time loop, where there is no PDE at all.

Questions

1 – 2 · A NETWORK FOR THE FUEL CHANNEL

1 **How can a network learn the steam fraction along the channel of the cell for every cell voltage?**

The network takes the position along the channel and the cell voltage, and is trained over 1.10 to 1.40 V until the species equation holds at points along the channel, with the local current set by the gap between the cell voltage and the local Nernst potential. The steam fraction at the inlet is built into the form of the solution, and the zero slope at the outlet is held by a loss term. After one training, any voltage in the range is read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

2 **Why does the local current make the fuel channel harder to solve, and how is the network's answer checked?**

The flux at the electrode is no longer given: the current at each point follows the local steam fraction, so the equation is nonlinear in it and needs a condition at the outlet, and it can no longer be marched from the inlet. The network takes the nonlinear term into its residual with no change of method. Its answer is scored against a boundary-value solver at a tight tolerance: at 1.29 V within 0.0005 of the steam fraction, where finite volumes on 50 cells are within 0.0042. *Ex_10.2 sections 3 – 5*

3 – 4 · THE BEST VOLTAGE

3 **How can the best cell voltage be found with a network, and what does the gradient save?**

The worth of an hour, the hydrogen made at its price less the electricity used at its price, is computed from the network's steam fraction and current. With the voltage as an input, automatic differentiation gives the derivative of the worth with respect to it, and gradient ascent finds 1.30 V in 200 steps and 0.2 s; a sweep of 61 runs of the reference solver finds the same voltage in 4 s. The prices are assumptions, so the voltage found is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

4 **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of the voltage and the temperature, because what produces more now ages the cell faster. A solver gives the worth, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives the gradient with respect to every input by automatic differentiation. Degradation rates are uncertain by an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

NEXT

L11.1 — Vision-Based Navigation

A network in a real-time control loop on a robot car, where the model is learned from camera frames and there is no PDE at all.

BEFORE THE NEXT SESSION

read	Liu ch. 5
run	Ex_10.2, sections 1 to 6
bring	your best voltage, and its worth
think about	what was physics, what was price

Block ten is finished: lithium in a particle, and steam in a ceramic cell - the course's last partial differential equations.